STORMWATER QUALITY MODELING IMPROVEMENTS NEEDED FOR SWMM

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PRESENTATION OVERVIEW

I Introduction

II Review of Existing SWMM Water Quality Formulations

III Problems Associated with SWMM Water Quality Formulations

IV Recommended SWMM Water Quality Modeling Enhancements

INTRODUCTION

- Version 1.0 of SWMM was released in July 1971
- Significant improvements and enhancements have been made to its capabilities in water quantity simulation
- Very few improvements have been made to its capabilities in water quality simulation

INTRODUCTION (cont)

- **Need for SWMM water quality modeling improvements are greater now than ever before**
  - **NPDES Phase II Implementation**
  - **TMDL Program**
  - **ESA Listing of Salmon in Northwest US**
  - **Increased interest and regulation throughout the developed world**

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REVIEW OF EXISTING SWMM WATER QUALITY FORMULATIONS

A. Pollutant Buildup on Land Surfaces (Accumulation)

B. Pollutant Washoff

C. Street Cleaning

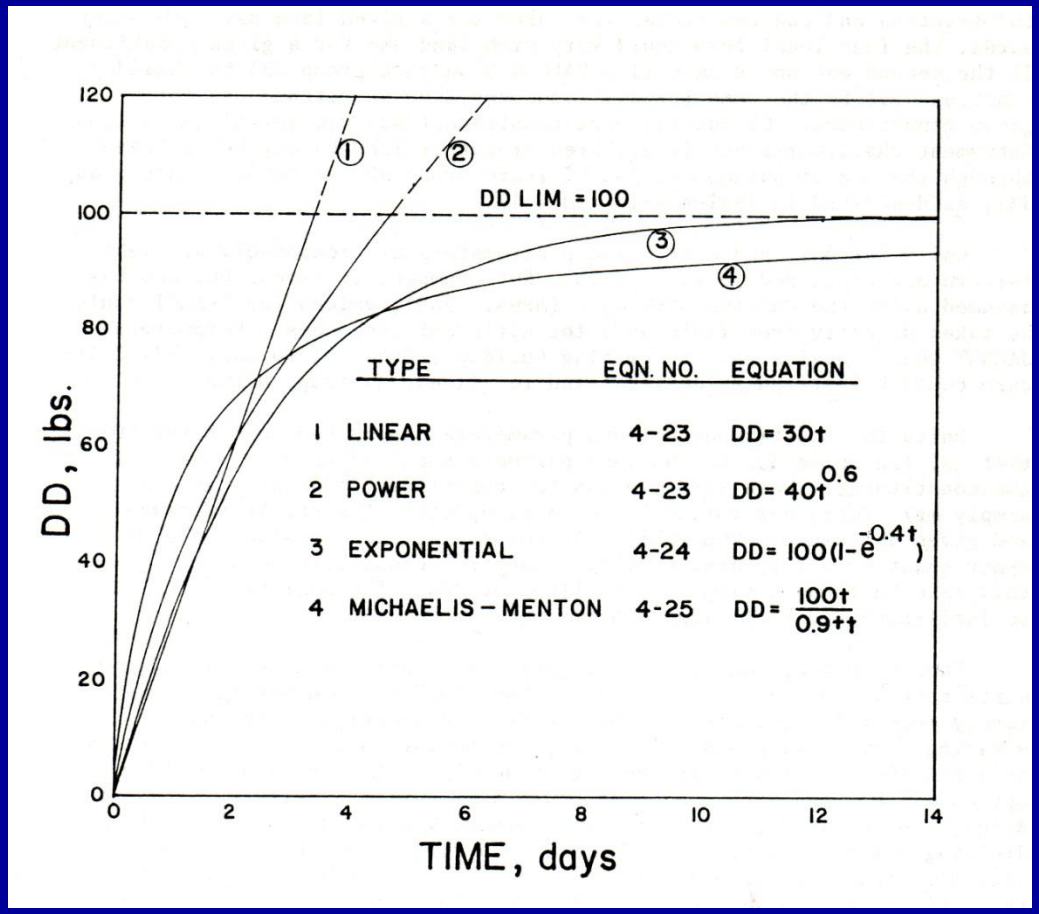
D. Sediment Trapping Catchbasins (STCBs)

A. Pollutant Buildup on Land Surfaces (Accumulation)

- **Occurs throughout a subcatchment area**
- **Magnitude of accumulation is a function of length of antecedent dry weather preceding the storm (*i.e.*, DRYDAYS)**
- **Several different buildup equations are available for use (*i.e.*, linear, power, exponential, Michaelis-Menton)**

A. Pollutant Buildup on Land Surfaces (cont)

Buildup Equations Available in SWMM



from Huber, et. al,, 1981

B. Pollutant Washoff

Two methods are available.

One uses the Exponential Washoff Equation:

$$PSHED(t) = PSHED_0 e^{-kt}$$

where PSHED(t) = quantity of pollutant on surface at time, t

$PSHED_0$ = initial quantity of pollutant

k = coefficient

$$K = RCOEF * r$$

where RCOEF = washoff coefficient, in^{-1}

r = runoff rate, in/hr

B. Washoff (cont)

Allen Burdoin (author of the washoff equation) assumed that 0.5 inches of rainfall in one hour would washoff 90% of the available pollutants.

- **Hence RCOEF = 4.6**
- **Availability factor created to reduce the magnitude of washoff**

$$AV = a + br^c$$

where AV = availability factor

a, b and c = coefficients

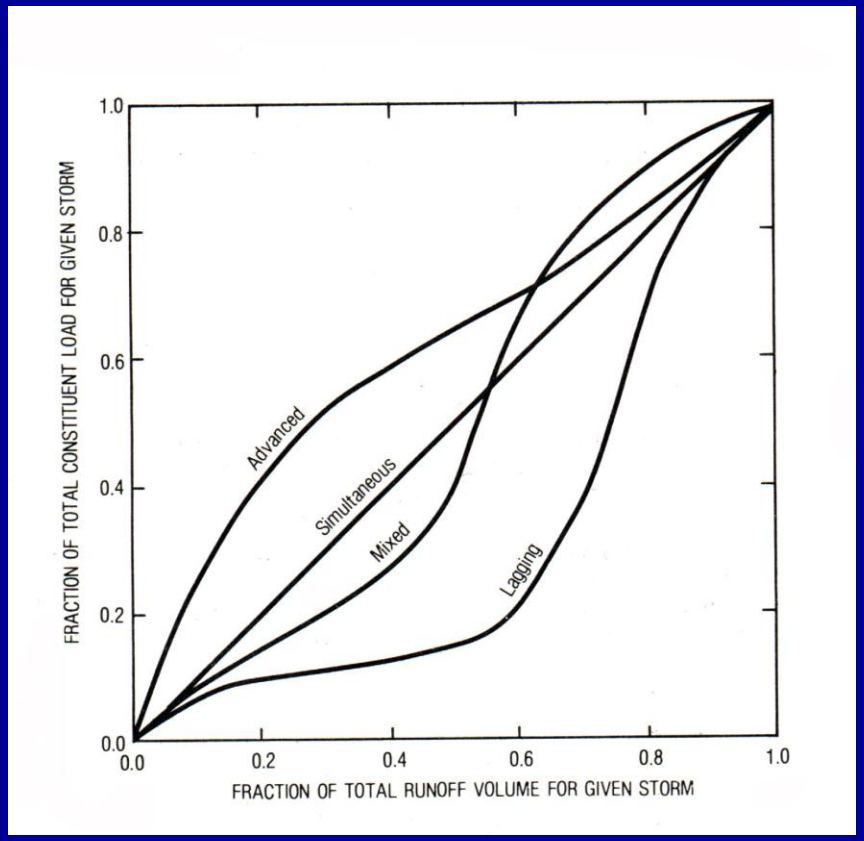
B. Washoff (cont)

Exponential equation always simulated a decreasing pollutant concentration over time independent of the runoff rate.

- **Always estimates a first flush**
- **High runoff rates in the middle of a storm would not produce higher concentrations which were being occasionally observed**

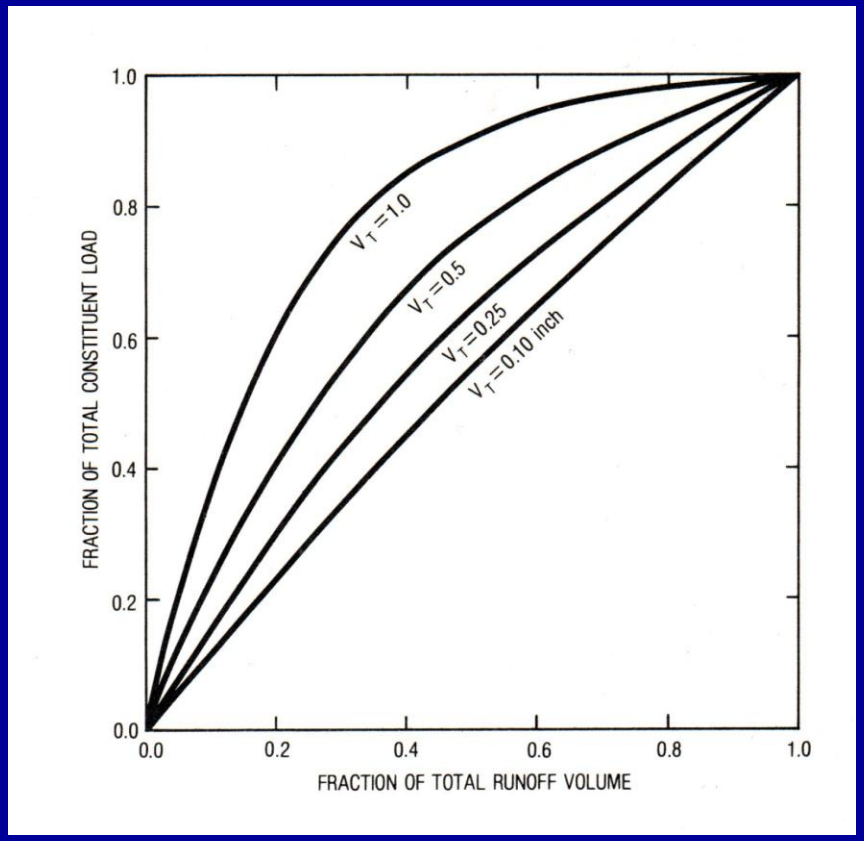
B. Washoff (cont)

Observed Load Characteristic Curves and Those Predicted by SWMM with RCOEF = 4.6



Load characteristic curves for advanced, simultaneous, lagging, and mixed types of concentration distributions.

from Alley, 1980



Predicted load characteristic curves for various storm runoff volumes (RCOEF=4.6)

B. Washoff (cont)

Version 3.0 of SWMM (Huber, et al, 1981) introduced a fix to the concentration problem by making the pollutant washoff proportional to runoff rate raised to a power called WASHPO.

$$POFF(t) = RCOEFX * r^{WASHPO} * PSHED(t)$$

where POFF(t) = pollutant quantity washoff at time, t, lb/sec (mg/sec)

RCOEFX = washoff coefficient
WASHPO = washoff coefficient

Concentration is now proportional to $r^{WASHPO-1}$.

AV is no longer used and two washoff parameters RCOEFX and WASHPO are now needed.

B. Washoff (cont)

The second washoff method is:

$$POFF(t) = R_c * Q(t)^n$$

where POFF(t) = pollutant quantity washoff at time, t

R_c, n = coefficients, vary by pollutant

Q(t) = flow rate at time, t

Washoff is totally independent of the accumulated pollutant and may be linked by limiting washoff to $PSHED_0$ (initial amount at start of the storm).

C. Street Cleaning

Occurs when the specified frequency of cleaning is equal to or greater than dry days between runoff events.

Availability factors (fractions) were created to reduce the amount of accumulated pollutants on the surface (PSHED) that is available to be removed by cleaning as follows:

$$REMAIN = 1.0 - AVSWP(J) * REFF(P)$$

where REMAIN = fraction of pollutant remaining on the surface after cleaning

AVSWP(J) = availability factor for land use J, and

REFF(P) = removal efficiency for pollutant P

D. Sediment Trapping Catchbasins (STCBs)

The pollutant removal effectiveness of STCBs is not modeled in SWMM.

The potential for a first flush of trapped catchbasin material can be modeled in SWMM.

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PROBLEMS ASSOCIATED WITH SWMM WATER QUALITY FORMULATIONS

A. Pollutant Buildup on Land Surfaces (Accumulation)

B. Pollutant Washoff

C. Street Cleaning

D. Sediment Trapping Catchbasins (STCBs)

A. Pollutant Buildup on Land Surface (Accumulation)

No spatial variability

- **impervious vs pervious surfaces**
- **different source areas (rooftops, driveways, streets)**

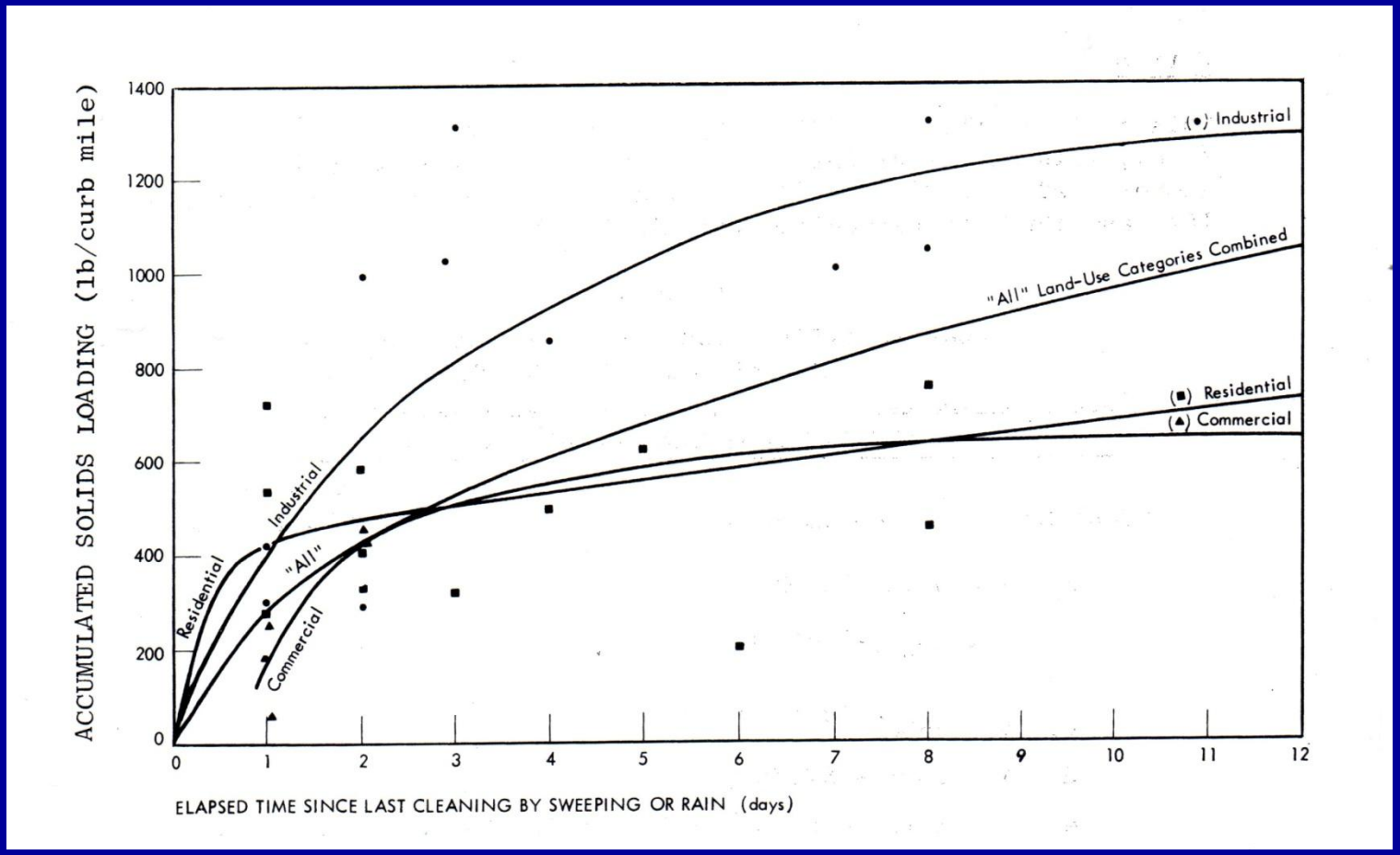
No wet weather "washon"

- **deposition of eroded materials from upland areas**
- **accumulation sometimes greater following storm**

Accumulation <u>always</u> increases with dry days.

A. Pollutant Buildup on Land Surface (cont)

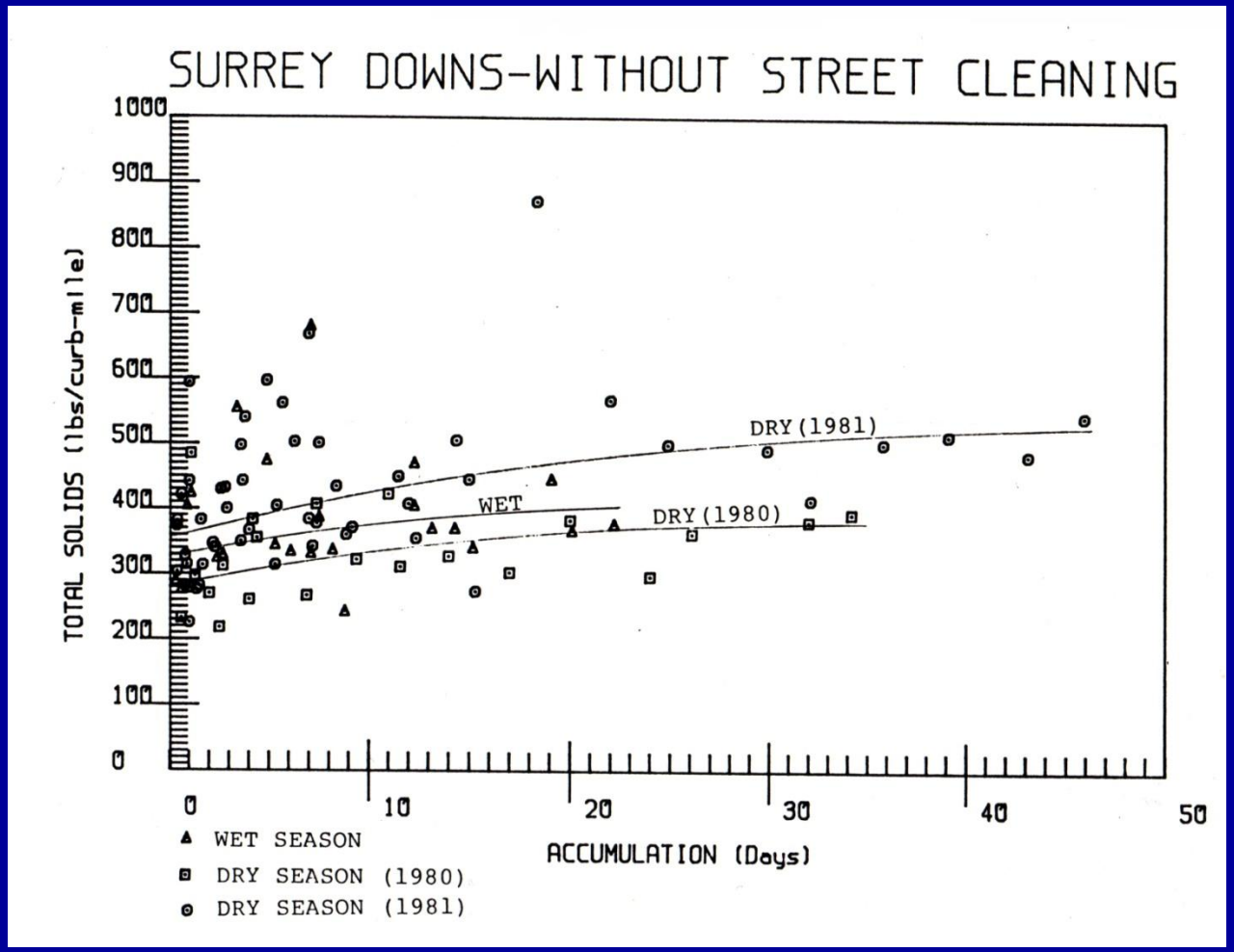
Accumulated Solids vs Dry Days by Land Use



from Sartor and Boyd, 1972

A. Pollutant Buildup on Land Surface (cont)

Accumulated Solids vs Dry Days



from Bellevue NURP (Pitt, 1985)

B. Pollutant Washoff

No spatial variability

- **impervious vs pervious surfaces**
- **different source areas**

No physical/chemical characteristics accounting for accumulation

- **particle size**
- **pollutant potency by particle size**
- **dissolved vs particulate forms**

No understanding of sediment transport mechanisms

- **entrainment/erosion/deposition**
- **armoring**

B. Pollutant Washoff (cont)

Washoff coefficients for exponential equation vary dramatically from storm to storm

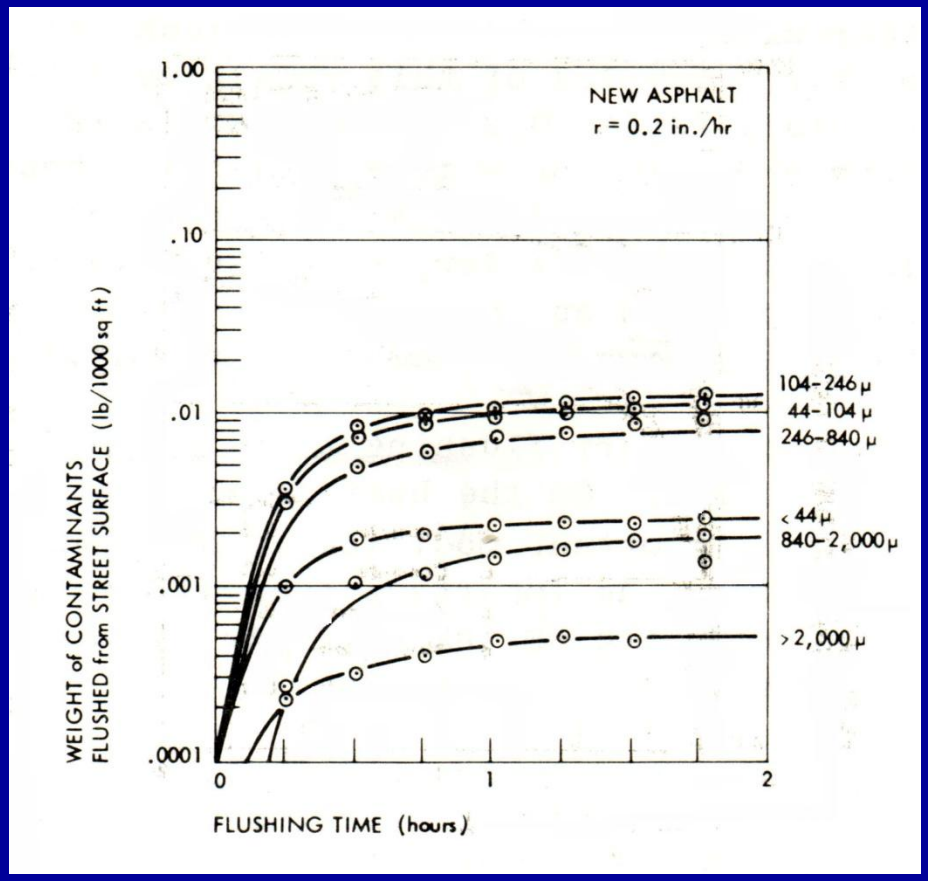
- **SWMM can not vary them**
- **Function of initial particle size distribution**

Availability factors are necessary

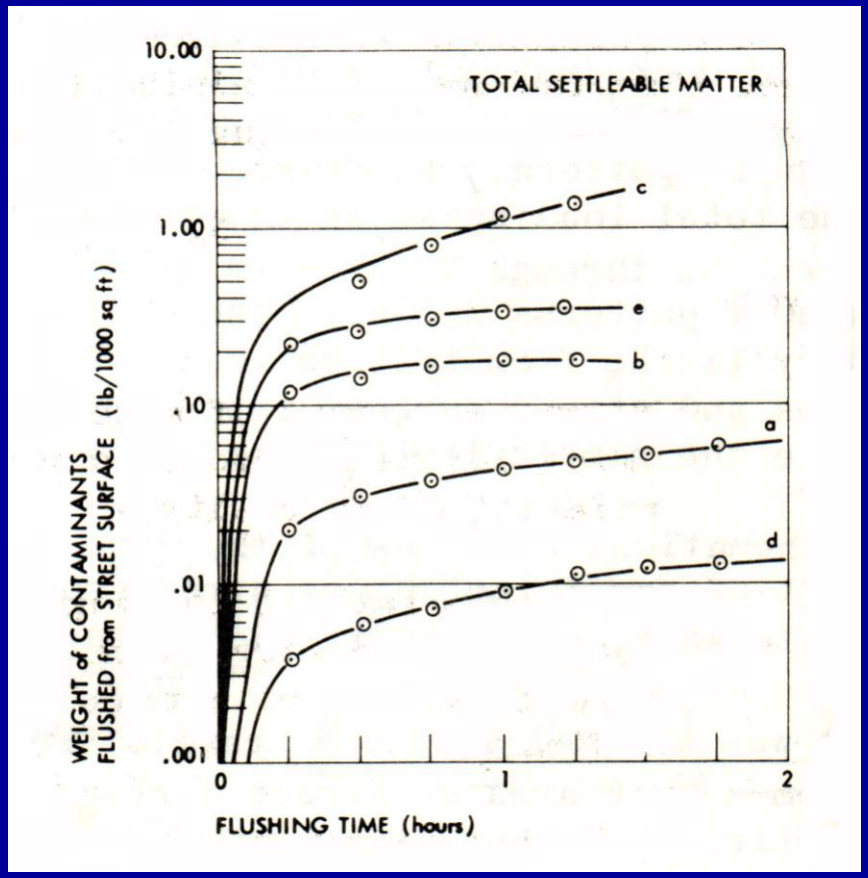
- **more complex than assumed**
- **function of surface conditions, rainfall intensity and location of accumulation in relationship to the upland stormwater flow paths**

B. Pollutant Washoff (cont)

Washoff Tests in Bakersfield, California



Variation by Particle Size



Variation by Street Roughness and Rainfall Intensity

from Sartor & Boyd, 1972

C. Street Cleaning

Effectiveness of street cleaner sediment pick up varies by:

- **initial amount and particle size distribution**
- **surface conditions**
- **type of equipment**
- **operational characteristics**

None of these parameters are considered in SWMM.

Effectiveness is assumed, <u>not</u> simulated.

D. Sediment Trapping Catchbasins (STCBs)

Use and periodic cleaning of STCBs, sedimentation manholes and vaults can provide reasonable pollutant reduction benefits that are worth considering and modeling.

Negative impacts of scour and downstream transport of sediments accumulated in STCBs are:

- **not that significant**
- **easily corrected by a periodic cleaning program**
- **could be explicitly modeled**

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RECOMMENDED SWMM WATER QUALITY MODELING ENHANCEMENTS

A. General Provisions

B. Sediment Accumulation on Impervious Surfaces

C. Sediment Washoff

D. Street Cleaning

E. Sediment Trapping Catchbasins (STCBs)

A. General Provisions

Increase the spatial variability of the accumulation/washoff/BMP processes

- **include both impervious and pervious areas**
- **provide for several different impervious source areas**
- **provide for flow and sediment transport linkages between these areas (*i.e.*, upstream to downstream)**

A. General Provisions (cont)

On each of the various source areas provide for:

- **sediment accumulation and accounting by particle size ranges**
- **sediment washoff by physical sediment transport equations**
- **BMP (street and STCB cleaning) interaction that estimates sediment removal based on the conditions that are assumed to exist when the BMP is assumed to occur**
- **simulate other pollutant washoff based on potency factors (both dissolved and particulate) that vary by particle size range**

A. General Provisions (cont)

Rely on the body of work and computer code largely completed by Sutherland and Jelen (et. al.) and Pitt and Voorhees (et. al.) in their respective development of computer programs, SIMPTM and SLAMM.

Rely on the body of work largely completed by Sansalone (et. al.) on the accumulation, transport and fate of dissolved pollutants in the urban environment.

B. Sediment Accumulation on Impervious Surfaces

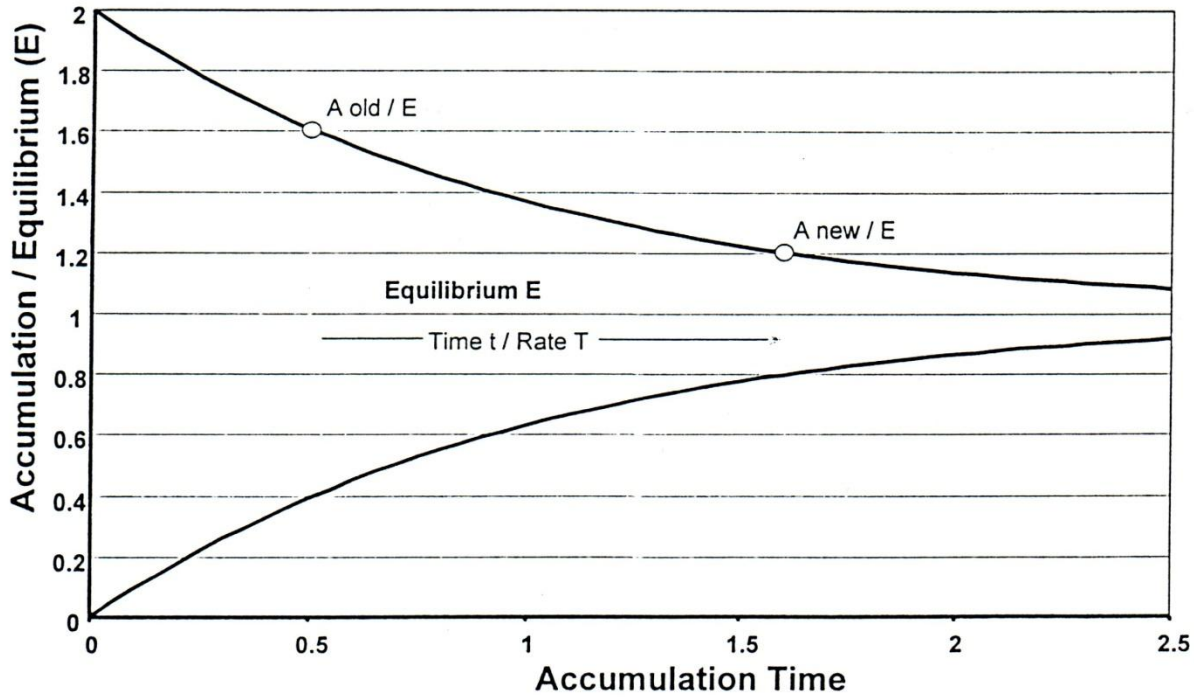
Use non-linear (*i.e.*, exponential) sediment accumulation and degradation equations with an equilibrium value not a maximum value.

Include wet weather "washon" that varies by rainfall depth and intensity.

Establish availability factors for washoff and BMP removal that are related to where the sediment accumulates in relationship to the upland stormwater flow pathways and how the BMP works.

B. Sediment Accumulation on Impervious Surfaces

Dry Weather Accumulation Function in SIMPTM



$$A_{new} = (A - E)(e^{-t/T} - 1)$$

from Sutherland and Jelen, 1996

C. Sediment Washoff

For impervious areas, use sediment transport based equations (same as those in SIMPTM) that consider:

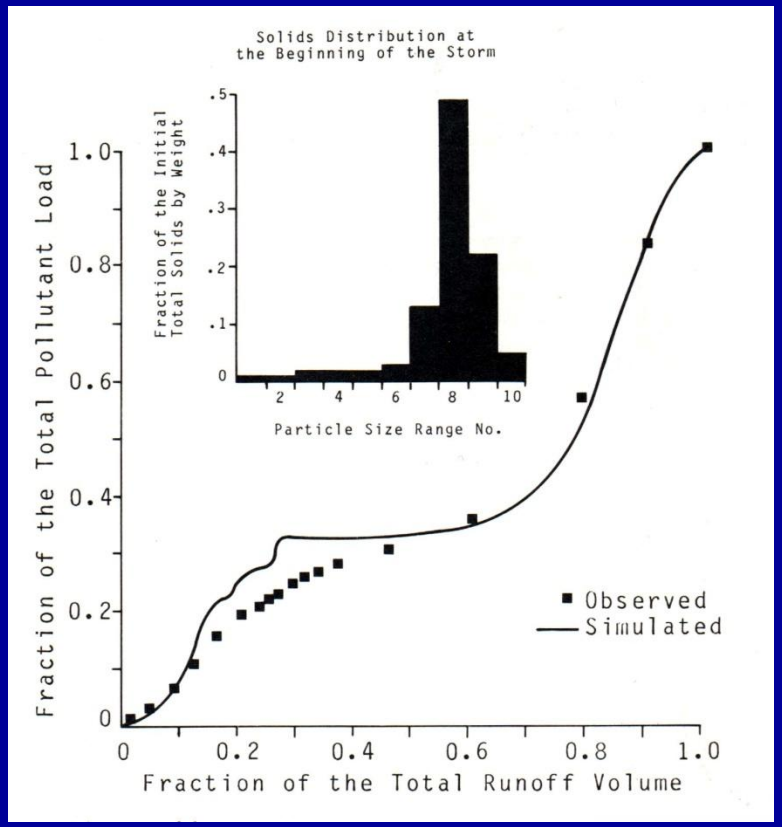
- **bed armoring**
- **particle size ranges**
- **dissolved fraction (*i.e.*, soluable transport)**

For pervious areas, use the same sediment transport equations but develop availability factors that are a function of:

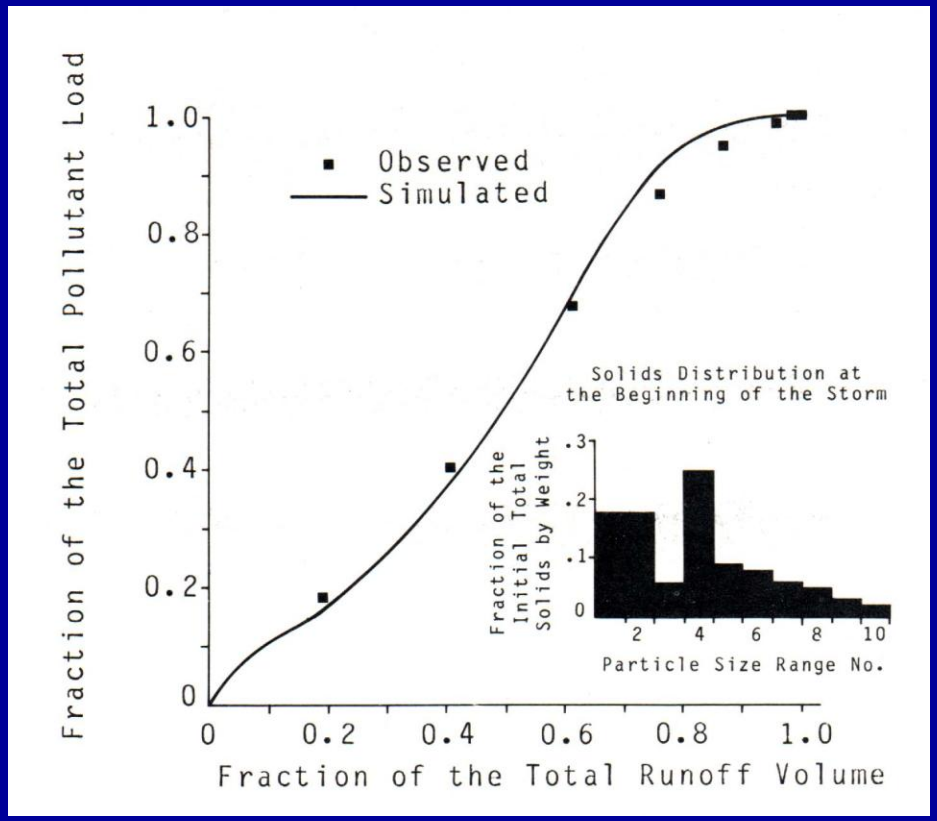
- **detachment of soil by rainfall intensity and runoff itself**
- **cover conditions that protect from detachment**

C. Sediment Washoff (cont)

Observed Load Characteristic Curves and Those Predicted by PTM



Calibration Results for Ross Ade Storm of 1 Nov 1972



Calibration Results for Livonia Drain Storm of 29 June 1977

from Ellis & Sutherland, 1979

D. Street Cleaning

Use the street cleaning equations (same as SIMPTM) that simulate street cleaner pickup as a function of sediment accumulation by particle size (see Chapter 9, Monograph 5 [1997], James Toronto Conference).

Allow user to specify exact dates of cleaning as an alternative to a specified frequency of cleaning.

Include an availability factor for parked car interference.

Street cleaning effectiveness in reducing sediment and associated pollutant washoff becomes <u>output</u> of the model, not an <u>input</u> to the model.

E. Sediment Trapping Catchbasins (STCBs)

Include the simulation of STCBs, manholes, or vaults and their ability to trap sediments and associated pollutants (use SIMPTM equations).

Include the simulation of cleaning operations at either a specified frequency or specific set of dates.

Include the ability to simulate the diversion of higher flows so scour can be reduced.

CONCLUSION

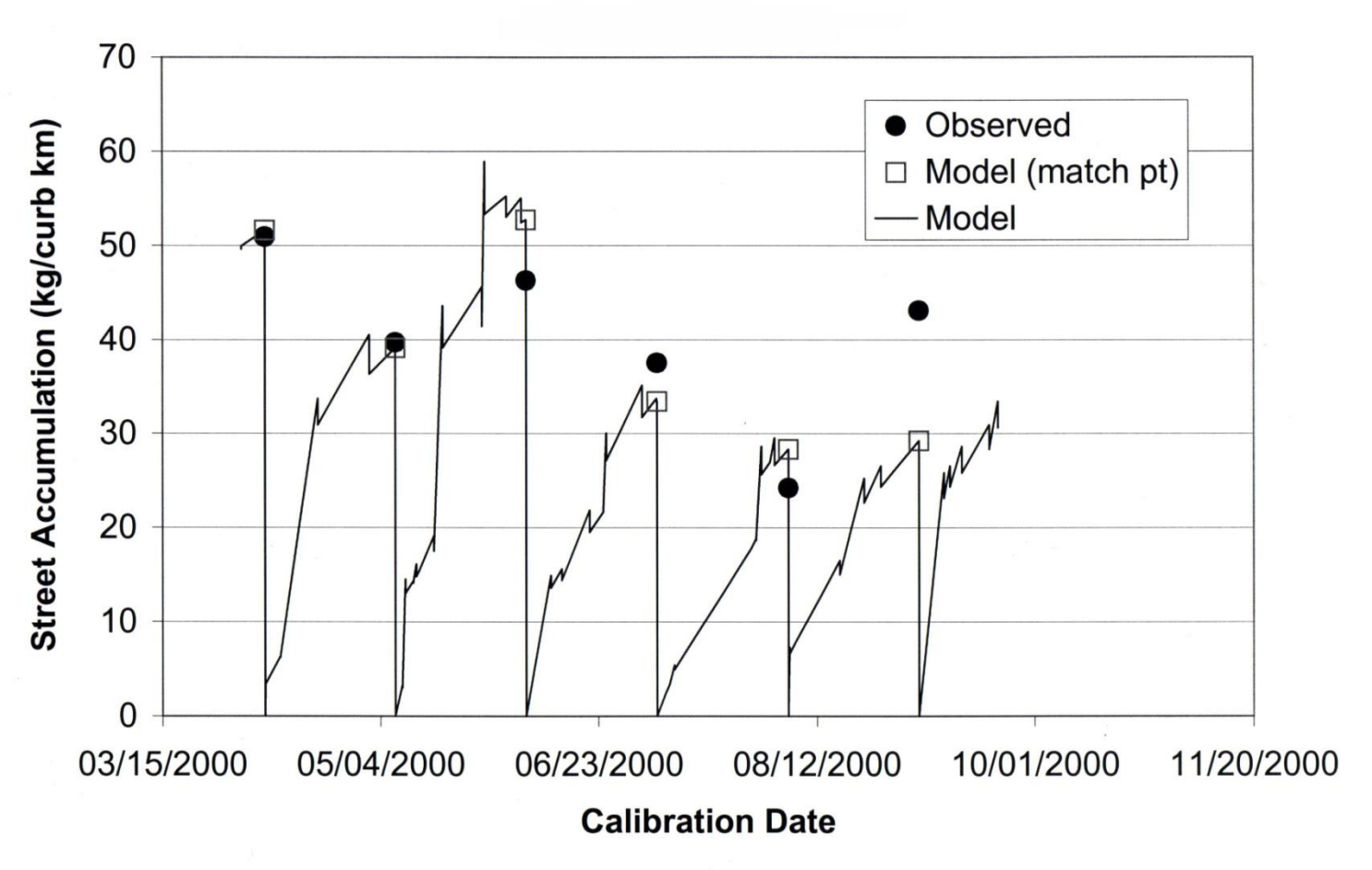
Once these recommended modeling improvements are implemented, SWMM will have the ability to <u>accurately</u> simulate:

- **the accumulation and washoff of urban sediments and other pollutants**
- **the pollutant reduction effectiveness of street and catchbasin cleaning**

The benefits of completing this undertaking will far exceed the cost.

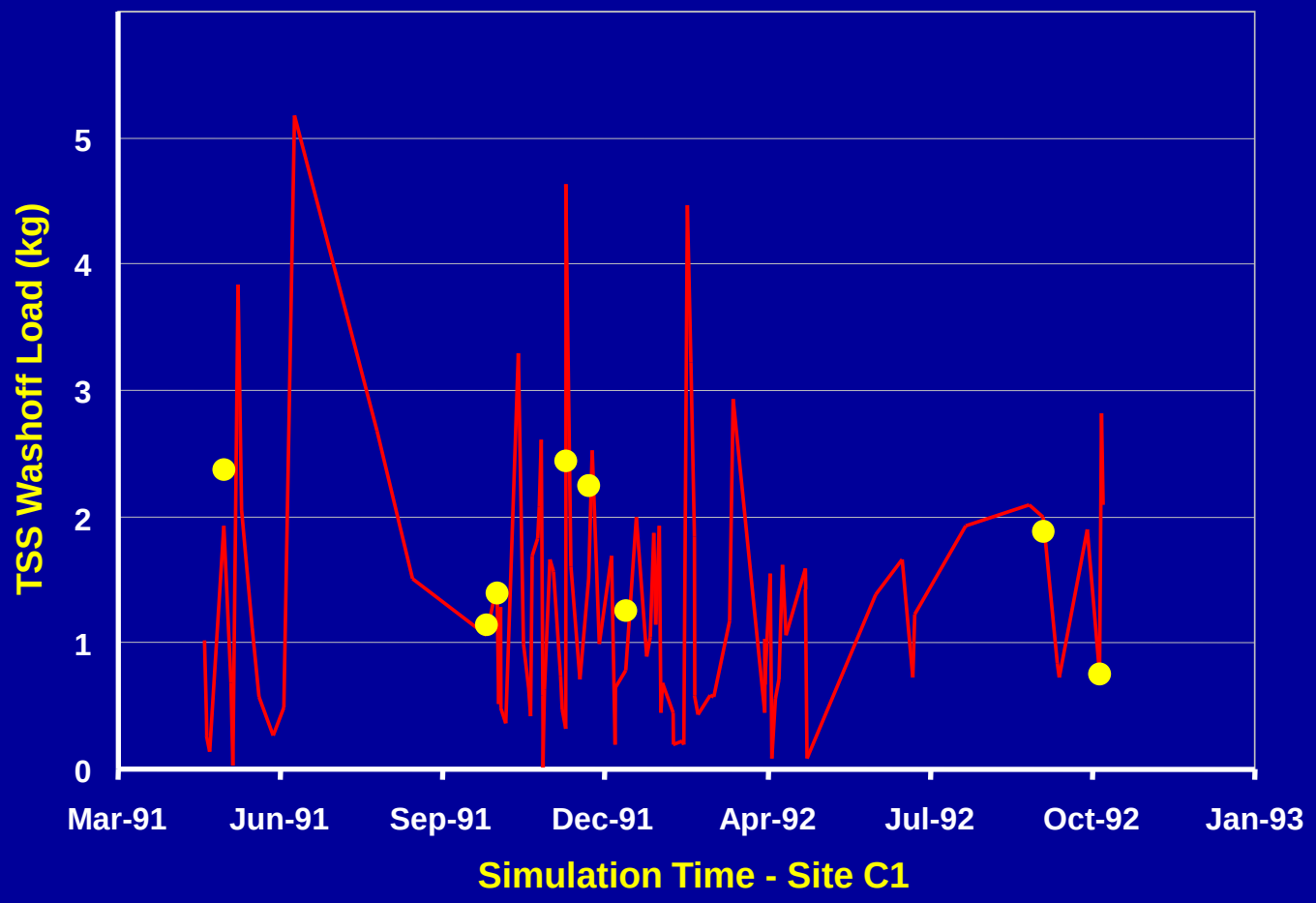
Observed vs Simulated Street Dirt Accumulation

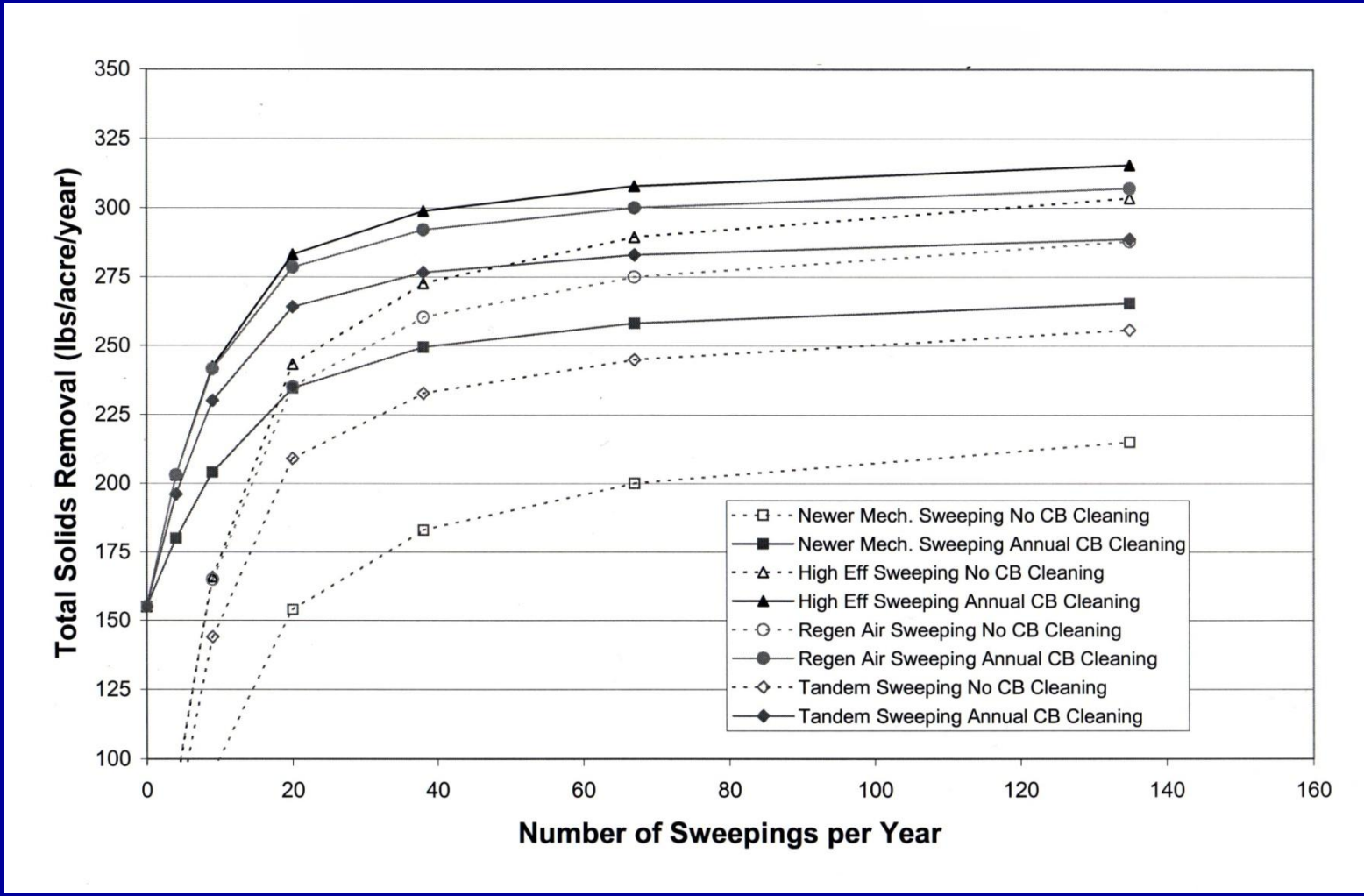
Single-Family Residential Site in Jackson, Michigan



Observed vs Simulated TSS Washoff

Downtown Commercial Site in Portland, Oregon



BMP Production Functions

Site #4 Parnell Highway in Jackson, Michigan